Hankuk University
PHY 101 General Physics I
Credits: 3 (lecture) + 1 (laboratory)

Course Description
Calculus-based introduction to classical mechanics for science and engineering students.

Learning Outcomes
1. Describe motion in one and two dimensions using vectors and calculus.
2. Apply Newton's laws of motion with free-body diagrams.
3. Use work-energy and conservation of momentum to solve problems.
4. Analyze rotational motion and simple harmonic motion.

Textbook
Halliday, Resnick and Walker, Fundamentals of Physics, 11th edition

Weekly Schedule
Week 1: Measurement, vectors
Week 2: Motion along a straight line
Week 3: Motion in two and three dimensions, projectile motion
Week 4: Newton's laws of motion
Week 5: Friction, circular motion
Week 6: Kinetic energy and work
Week 7: Potential energy and conservation of energy
Week 8: Midterm exam
Week 9: Center of mass and linear momentum, collisions
Week 10: Rotation, torque and moment of inertia
Week 11: Rolling, angular momentum
Week 12: Equilibrium and elasticity
Week 13: Gravitation
Week 14: Oscillations and simple harmonic motion
Week 15: Final exam

Assessment
Homework 20%, Laboratory 15%, Midterm 30%, Final exam 35%